

Xiang Li 李翔

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About me

I am a first-year Ph.D. student in a joint training programme between **BNU–HKBU United International College** and **Great Bay University**. I received my M.Sc. in Biological Engineering from the **University of Saskatchewan** (2026), where I worked with Prof. Yen-Han Lin on machine learning approaches for redox potential–controlled very-high-gravity ethanol fermentation, and my B.Eng. in Biological Engineering from **Northwest A&F University** (2024).

My research sits where **fermentation bioprocess engineering** meets **machine learning**. I build soft sensors and kinetic models that turn cheap, continuous online signals — oxidation–reduction potential (ORP) in particular — into quantitative, forward-looking estimates of metabolic activity, so that a fermentation can be steered before it fails rather than analysed after it ends.

Position: Ph.D. Student, Year 1
Programme: BNU–HKBU UIC × Great Bay University
M.Sc.: University of Saskatchewan, 2026 (GPA 4.0/4.0)
B.Eng.: Northwest A&F University, 2024

I am open to research collaborations and discussions on machine learning for bioprocess monitoring, kinetic modelling, and fermentation process control.

Research interests 研究方向

Fermentation process control; Kinetic modelling & parameter inference; Deep learning for bioprocesses (AI4Bio);
Soft sensing & state estimation; Bayesian uncertainty quantification; Bioenergy & cleaner biomanufacturing

Publications 学术成果

Bayesian inference on fermentation kinetics: Comparative analysis with frequentist approach

PUBLISHED

Xiang Li, Yen-Han Lin

Bioresource Technology, 2026, 444, 134012. IF 9.0, JCR Q1.

DOI → (https://doi.org/10.1016/j.biortech.2026.134012) ScienceDirect → (https://www.sciencedirect.com/science/article/pii/S0960852426000933)

Enhancing cleaner biomanufacturing of ethanol: A temporal deep learning soft sensor for ORP-driven metabolic forecasting in very-high-gravity fermentation

UNDER REVIEW

Xiang Li, Yen-Han Lin

Journal of Cleaner Production — under review, 2026.

Modified Hyperbolic Secant (MHS) Models for Fermentation Kinetics and Uncertainty

POSTER

Xiang Li, Yen-Han Lin

Poster, YABEC 2026 (Asian Young Biotechnologists Network), University of Saskatchewan.

CV 简历

EDUCATION

Ph.D. Student, Biological Engineering 2026 – present

BNU–HKBU United International College & Great Bay University, Guangdong, China
Joint training programme. Year 1.

M.Sc., Biological Engineering 2025 – 2026

University of Saskatchewan, Saskatoon, Canada
GPA 4.0/4.0. Supervisor: Prof. Yen-Han Lin. Thesis: Machine learning approaches for redox potential-controlled very high gravity ethanol fermentation.

B.Eng., Biological Engineering 2020 – 2024

Northwest A&F University, Yangling, China
GPA 3.71/4.0, rank 3/63. Thesis: Deep learning-based image processing of maize allele-specific chromatin interactions and regulation of imprinted genes.

RESEARCH EXPERIENCE

Deep learning monitoring and forecasting system for ethanol fermentation

Jun. 2025 – present

- Fused sparse offline measurements with dense online ORP signals to build a high-resolution training set; developed a real-time state-estimation model (ORPSN, RNN) and a multi-horizon forecasting model (ORPSC, LSTM encoder–decoder) for metabolic dynamics.
- **Outcome:** Achieved high-accuracy soft sensing of ethanol production and glucose consumption rates; maintained accuracy over an 8-hour forecast horizon, providing a basis for proactive feed-forward control.

Bayesian inference for fermentation kinetics and uncertainty analysis

Jan. 2025 – Dec. 2025

- Built Bayesian hierarchical kinetic models for multiple fermentation systems with algorithmic and literature-informed priors; systematically compared Bayesian and frequentist inference for parameter identifiability and predictive stability under sparse data.
- **Outcome:** Resolved non-convergence of conventional methods in small-sample settings, giving robust kinetic parameter estimates with explicitly quantified uncertainty and improved biological interpretability.

Allele-specific chromatin interactions and imprinted gene regulation in maize endosperm

Mar. 2023 – May 2024

- Applied autoencoders and U-Net to denoise and enhance the resolution of Hi-C data from reciprocal crosses of maize inbred lines B73 and Mo17; constructed allele-specific chromatin interaction maps; partitioned A/B compartments and identified TADs genome-wide; analysed compartment distributions around imprinted genes.
- **Outcome:** Built an allele-resolved 3D chromatin architecture of maize endosperm and showed that the compartmental distribution of imprinted genes has no clear parent-of-origin dependence, indicating that chromatin reorganisation at the A/B compartment level is not the principal mechanism of imprinting regulation there.

Metagenomics and culturomics for tobacco biological control

Nov. 2021 – Nov. 2022

- Screened and validated antagonistic strains and assembled consortia; designed and ran field trials; performed 16S rRNA amplicon analysis of the tobacco rhizosphere microbiome.
- **Outcome:** Established a biocontrol strain library with good field efficacy and developed an effective composite microbial agent.

RESEARCH VISITS & INDUSTRY

Visiting Student

Jan. 2026 – present

Shanghai Jiao Tong University, School of Life Sciences and Biotechnology

Project: deep learning–based metabolic modelling of filamentous fungi and its industrial application.

Sales Engineer

Sep. 2024 – Dec. 2024

Servicebio Technology Co., Ltd., Zhengzhou, China

Technical support and product application for life-science research reagents.

HONOURS & AWARDS

University of Saskatchewan CGPS 75th Recruitment Scholarship	Jan. 2025
National Scholarship for Encouragement (国家励志奖学金), Northwest A&F University	Nov. 2021
First Prize, National Undergraduate Life Sciences Competition (Innovation & Entrepreneurship)	Nov. 2021
Meritorious Winner (top 8%), Mathematical Contest in Modeling, COMAP, USA	Apr. 2021
First-Class Academic Scholarship, College of Life Sciences, Northwest A&F University	Dec. 2020

SKILLS

Python; PyTorch; TensorFlow / Keras; NumPy & pandas; Bayesian inference (MCMC); Kinetic modelling;
Bioreactor operation & ORP control; HPLC analysis; R; Git